Supplemental materials (SM2) for: An assessment of biodiversity data shortfalls for ectomycorrhizal fungi in Canada

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1 Decisions log

Purpose
This log inventories the decisions **we** made in assembling and analyzing the dataset (Tables A1–A6), and, separately, the upstream decisions **outside our control** that are baked into the external datasets we rely on (Tables B1 - B9).

How to read the tables
Each row is one decision. The *Choice* column records what we did; the *Rationale* column records the stated reason

where one exists in the code or Supplemental Materials. Parameter values in `code font` are taken directly from the pipeline scripts; the parenthetical script reference (e.g. `02_globalfungi.R`) points to the source.

1.1 PART A — Decisions made by us

1.1.1 Table A1. GlobalFungi record assembly (`02_globalfungi.R`)

#	Decision	Choice	Rationale
A1	Barcoding-region filter	Retain `barcoding_region %in% {ITS2, ITSboth}`	Restricts to samples carrying an ITS2 read, the region used for SH assignment; sets up parity with the GenBank ITS2 restriction (Detailed methods SM1).
A2	Manipulation filter	Retain `manipulated == "NO"`	Exclude experimentally manipulated samples so records reflect natural communities (Detailed methods SM1).
A3	Substrate exclusion	Drop substrate in `{shoot, air, water, sediment}`	These substrates are irrelevant to (soil/root-associated) EcM fungi (Detailed methods SM1).

#	Decision	Choice	Rationale
A4	Species-level GlobalFungi matrix	**Not used**; work from the SH-level matrix only, decoded through the pinned UNITE build	Keeps a single, self-consistent SH definition across the project (`data_raw/DATA-DICTIONARY.md`).

1.1.2 Table A2. GenBank record assembly (`03_genbank.R`)

#	Decision	Choice	Rationale
B1	Search query	`"Canada"[Country] AND "Fungi"[Organism] AND ("internal transcribed spacer"[All Fields] OR "ITS"[All Fields])`; **no** date cutoff, **no** lifestyle keyword	Returns all matching fungal ITS records regardless of guild; we apply EcM filtering later and consistently (Detailed methods SM1).
B2	Sub-region extraction	ITSx (HMMER backend, Fungi profile set) to detect/extract ITS1 and ITS2	GenBank-side analogue of GlobalFungi's ITS2/ITSboth restriction, for cross-source comparability (Detailed methods SM1).
B3	ITS2 minimum length	Exclude ITS2 fragment `< 100 bp`	Floor to exclude spurious very short calls that trivially tie with many unrelated UNITE references (`03_genbank.R`, Detailed methods SM1).

#	Decision	Choice	Rationale
B4	SH matching input	Match on the **extracted ITS2 fragment only**; ITS1 not independently classified	Deliberate simplification preserving a one-row-per-accession schema; we disclose the consequence (loss of exact sub-region parity with GlobalFungi) and diagnose it via `genbank_its1_resolution_crosstab.csv` (Detailed methods SM1).
B5	`vsearch` identity threshold	`usearch_global` at **98.5% identity**	Matches GlobalFungi's documented SH-assignment criterion (Detailed methods SM1).
B6	Tie resolution — named vs dark	Named-species hits compared by species name; dark hits (`_sp` placeholders) compared by SH code	Follows UNITE's own dark-taxon convention (Nilsson et al. (2019); Ryberg and Nilsson (2018)) (Detailed methods SM1).
B7	Tie resolution — fallback rank	Agree at SH → assign SH; disagree but share genus → keep at **genus-only** (`sh_code`/`species` = NA); disagree at genus → **exclude** as ambiguous	"Climb to the lowest unambiguous rank" principle (Heeger et al. (2018)); excluded records logged to `genbank_ambiguous_species_excluded.csv` (Detailed methods SM1).

B8	Host taxon source	Structured `host` qualifier, with fallback regex extraction of binomials from `isolation_source`	`host` is the most reliable field; `isolation_source` recovers additional hosts (Detailed methods SM1).

1.1.3 Table A3. Host-name normalization — shared across sources (`canonicalize_host()`, `00_setup.R`)

#	Decision	Choice	Rationale
C1	Single shared normalizer	One `canonicalize_host()` applied identically wherever we match a raw host string to a species list (GenBank host field; Eltonian; Hutchinsonian)	Raw host names are free text (binomials, misspellings, abbreviations, common names, inconsistent case/punctuation); one shared function avoids ad hoc per-analysis cleaning and silent match failures (Detailed methods SM1).
C2	Normalization steps	Strip surrounding punctuation → collapse whitespace → keep first two tokens → strip per-token punctuation → apply curated typo lookup → case-normalize (Titlecase genus, lowercase epithet) → reject non-Latin-name patterns → NA	Reduce each raw string to a standardized “Genus species”/“Genus” form, or NA if unresolvable (`00_setup.R`).

#	Decision	Choice	Rationale
C3	Typo-correction lookup	Curated `typo_fixes` (currently `Abies balamifera → Abies balsamea`), extended as new misspellings surface	Manual, auditable corrections rather than fuzzy auto-correction (`00_setup.R`).

1.1.4 Table A4. Source combination, spatial validation, EcM guild, and site/occurrence grain conventions (`04_combine_ecm_dataset.R`, `00_setup.R`)

#	Decision	Choice	Rationale
D1	Coordinate validation	Spatial-containment check vs GADM Canada boundary, **buffered by 2 km** (in the Canada Albers Equal Area Conic projection) → logical `coord_in_canada` (TRUE / FALSE / NA); original coordinates always preserved	Distinguishes validated-inside from present-but-outside vs no-coordinates, without discarding data. The 2 km buffer absorbs GADM coastline-simplification and topology artefacts that would otherwise exclude a handful of genuinely Canadian coastal and island GlobalFungi records (Detailed methods SM1).

#	Decision	Choice	Rationale
D2	Spatial vs non-spatial use	Spatial analyses filter `coord_in_canada == TRUE`; non-spatial enumerations (richness, counts) include **all** records	For counts, we treat a GlobalFungi `country == Canada` label as authoritative even where coordinates fall marginally outside the buffered polygon (Detailed methods SM1).
D3	EcM guild assignment	Genus-level: retain records whose UNITE-assigned genus is `primary_lifestyle == "ectomycorrhizal"` in FungalTraits (genus lower-cased before matching)	Genus-level EcM status from a single authoritative trait source (Detailed methods SM1).
D4	FungalTraits version	**Pinned** to v1.2 (`FungalTraits_1-2.csv`); one file used by all scripts via `paths$fungaltraits`	Held fixed rather than tracked to a live source, so every script shares an identical EcM genus definition; provenance and SHA256 recorded in `fungaltraits_version.txt` (mirrors the UNITE build-pinning rationale given in Supplemental Materials SM1).

#	Decision	Choice	Rationale
D5	"Site" definition	Bin lat/lon to **3 decimal places** (~100 m); `add_site_id()`	Collapses GPS-precision noise across repeat readings at one location without merging distinct sites (`00_setup.R`, Detailed methods SM1).
D6	Occurrence grain (Wallacean)	Distinct **30 arc-second** (~1 km^2) grid cells; `snap_30arcsec()`	Aggregates occurrences to ~1 km^2 cells matching common SDM practice (`00_setup.R`, Detailed methods SM1).
D7	Where we apply 3-decimal site-binning	We use the `add_site_id()` 3-decimal site-binning (D5) for the ecozone sampling-coverage counts, but do **not** apply it to the Wallacean per-taxon location table, whose within-Canada occurrence counts use raw distinct coordinates; we use the separate **30 arc-second** occupancy grid (D6) for the worldwide occupancy comparison and Figure 2	Distinct raw coordinates preserve per-record occurrences for the Canada location counts, while the coarser 30 arc-second grid gives a like-for-like occupancy comparison against the worldwide GlobalFungi dataset (`11_wallacean.R`, Detailed methods SM1).

1.1.5 Table A5. Reference / host-data preparation (`05_prepare_fungalroot.R`, `06_host_species.R`, `07_bien2_ranges.R`, `08_host_rasters.R`)

#	Decision	Choice	Rationale
E1	Host determination — two routes	Species-level (FungalRoot occurrence records) **OR** genus-level (FungalRoot Table S2); combined with OR	Two independent routes; a species/genus can qualify via either or both (Detailed methods SM1).
E2	Qualifying species-level labels	Count as EcM-demonstrated only unambiguous EcM-positive labels: `EcM`, `EcM, no AM colonization`, `ErM,EcM`	Restrict to unambiguous positive evidence (Detailed methods SM1).
E3	Excluded label — `EcM, AM undetermined`	Excluded as qualifying evidence (but not disqualifying)	Denotes an EcM report with AM status unsurveyed, not a confirmed AM-negative; contradicted in a checked case (*Dryopteris filix-mas*) (Detailed methods SM1).
E4	Excluded label — `EcM,AM` (dual-positive)	Excluded as qualifying evidence	On inspection it was the sole EcM evidence for 118 of 698 candidate species across 20 genera (e.g. *Acer*, *Allium*, *Fraxinus*, *Juglans*) with no unambiguous record elsewhere (Detailed methods SM1).

#	Decision	Choice	Rationale
E5	Genus-level qualifying labels	Table S2 assignment of `EcM` or `EcM-AM` qualifies	`EcM-AM` captures genuine dual-mycorrhizal genera (*Salix*, *Populus*, *Eucalyptus*) the binary species rule cannot express; at least 67% species-consistency basis is FungalRoot's own (Detailed methods SM1).
E6	Non-standard Table S2 rows	Excluded by default (e.g. "species-specific: EcM or NM-AM")	Do not match either qualifying label (Detailed methods SM1).
E7	No literature supplementation	Do **not** add case-by-case literature EcM records (e.g. *Acer*, *Fraxinus*, *Juglans*)	Keep host determination reproducible from a single defined source; disclosed cost is under-counting a few genera (Detailed methods SM1).
E8	Native host list construction	BIEN species with Canadian distributions → filter to native via Native Status Resolver → retain species qualifying under **either** route in E1 (species-level occurrence list **OR** Table S2 genus list), combined with OR	Applies the two-route host determination (E1) to the native flora; no growth-form restriction. `evidence_source` records which route(s) applied (`06_host_species.R`, Step 4; Detailed methods SM1).

#	Decision	Choice	Rationale
E9	Growth-form assignment	Four-stage chain, applied in order: BIEN trait database → GIFT trait 1.2.1 (`agreement = 0.66`) → modal growth form of congeners already in the host list → curated manual override. The stage that supplied each value is recorded in `growth_form_source`	Growth form drives the tree / non-tree split reported in the main text and Figure 4, so it must be resolved for every host species; BIEN alone covers 215 of 272. Recording the source keeps the 42 congener-inferred values distinguishable from values taken from a trait record for the species itself (`06_host_species.R`, Steps 5–7c).

#	Decision	Choice	Rationale
E10	Unresolved synonymy	Accept drop-outs from independent taxonomic resolvers rather than harmonizing synonyms	A general synonym-harmonization step could shift counts for many species; we flag this as a known limitation instead. The genus route (E1) already rescues most such cases — e.g. we retain *Alnus alnobetula* via genus *Alnus* in Table S2 even though its species-level EcM evidence is filed under *A. crispa* — so a residual drop-out requires the genus to be absent from Table S2 as well (Detailed methods SM1).
E11	Host-range rasters	BIEN2 MaxEnt range polygons → reproject WGS84 → clip to Canada → rasterize to **0.5° grid**; three layers (host richness, host-data richness, host-data proportion)	Standardized grid for host richness and data-coverage comparison (Detailed methods SM1).

1.1.6 Table A6. Shortfall-analysis decisions (`09–21`)

#	Shortfall	Decision	Choice	Rationale
F1	Linnean	Richness units	Report unique UNITE SH codes, genera, and named species (excluding `_sp`)	SH codes are the primary unit: consistent OTU applicable to both sources, and many EcM taxa are unnamed (Detailed methods SM1).
F2	Linnean	Dark fraction	Proportion of detected SH codes lacking a species-level UNITE name (combined and per source)	Operational definition of the "dark fraction" (Detailed methods SM1).
F3	Linnean	van Galen comparison	Filter van Galen per-sample table to `Country == Canada` and `biome` `{forest, tundra}`; report **unweighted mean** of per-sample unassigned proportion, as a non-equivalent reference only	Subset most comparable to our sampling; explicitly flagged as non-equivalent (different taxonomic units: OTU vs SH) (Detailed methods SM1).

#	Shortfall	Decision	Choice	Rationale
F4	Linnean	GBIF vouchered complement	GBIF Kingdom Fungi, Canada, basis PRESERVED_SPECIMEN/LIVING_SPECIMEN, geospatial issues excluded, filtered to species-rank records in EcM genera (FungalTraits)	Vouchered specimen complement to sequence data, archived download key for reproducibility (retrieved 6 Mar 2026) (Detailed methods SM1).
F5	Wallacean	Occurrence levels	Count occurrences per taxon at SH, genus, and named-species levels	Multiple taxonomic resolutions of the shortfall (Detailed methods SM1).
F6	Wallacean	Reporting thresholds	Report proportion of taxa known from a single occurrence and from < 30 occurrences (Canada and global); tabulate common minimum-occurrence SDM thresholds	Straightforward shortfall measures; 30 is a common SDM minimum-sample rule of thumb (Detailed methods SM1).
F7	Wallacean	Data source	Occurrence/occupancy counts drawn from **GlobalFungi only**	GlobalFungi provides the structured sampling frame required for meaningful occupancy comparison (Detailed methods SM1).

#	Shortfall	Decision	Choice	Rationale
F8	Wallacean	Global comparator	Map each Canadian species/SH to all globally-known SH codes of that identity, matched against the full global GF v5 SH matrix under the same quality filters	Consistent global denominator (Detailed methods SM1).
F9	Prestonian	Source & filters	BioTIME (downloaded 5 Mar 2026): studies with at least 1 Canadian location, at least 2 time points, fungi-consistent organism type	Minimal criteria for a fungal time series in Canada (Detailed methods SM1).
F10	Prestonian	Shortfall metric	% of named Canadian EcM species lacking any abundance time series **anywhere on Earth**	Operational definition (Detailed methods SM1).
F11	Darwinian	Source	JGI MycoCosm "All Fungi" listing (retrieved 5 Mar 2026 by copying the page; bulk export required login and was non-functional)	Only complete accessible listing at the time; fungi-only page needs no extra filtering (Detailed methods SM1).

#	Shortfall	Decision	Choice	Rationale
F12	Darwinian	Matching	**Exact** set-membership match after normalization: genus = first token of MycoCosm `taxon_name`, species = first two tokens, both lower-cased/trimmed; matched against lower-cased `emf` genera and named species (placeholder epithets sp/spp/cf/aff excluded)	We use no fuzzy/edit-distance matching or tolerance (`15_darwinian.R`). Limitation: exact matching misses cross-database synonyms and spelling variants, which slightly *over*-states the shortfall.
F13	Darwinian	Shortfall metric	% of named Canadian EcM species absent from MycoCosm	Operational definition (Detailed methods SM1).

#	Shortfall	Decision	Choice	Rationale
F14	Raunkiæran	Traits assessed	Six FungalTraits columns: 3 morphological (fruitbody type, hymenium type, exploration type) + 3 ecological (secondary lifestyle, endophytic-interaction capability, host specificity)	Columns carrying information meaningful for EcM fungi (Detailed methods SM1).
F15	Raunkiæran	Excluded trait	Growth form excluded	"Filamentous mycelium" for all EcM genera — uninformative (Detailed methods SM1).
F16	Raunkiæran	Coverage definition	Proportion of genus × trait cells non-missing and not literal "unknown"; secondary lifestyle & endophytic capability treated as a **lower bound**	Blanks in those two traits are ambiguous between "not assessed" and "trait absent" (Detailed methods SM1).

#	Shortfall	Decision	Choice	Rationale
F17	Hutchinsonian	Ecozone unit	Canada's 15 terrestrial ecozones (National Ecological Framework); report number/proportion sampled across a range of site thresholds	Standard national ecozone framework; no ecozones excluded (all contain predicted host habitat) (Detailed methods SM1).
F18	Hutchinsonian	Offshore points	Points just offshore of the ecoregion coastline snapped to nearest ecozone within **5 km**	Recover coastal samples misplaced by coastline generalization; ecozone areas use the same polygon basis (Detailed methods SM1, Table S1).
F19	Hutchinsonian	Climate variables	WorldClim 2.1 BIO1 (MAT) and BIO12 (MAP)	Two axes defining Canadian climate space (Detailed methods SM1).
F20	Hutchinsonian	Climate grid	Aggregate WorldClim 30 to ~10 (`AGG_FACTOR = 20`)	Match the spatial grain of the source workflow (Hébert et al. (2026)) (`17_hutchinsonian.R`).

#	Shortfall	Decision	Choice	Rationale
F21	Hutchinsonian	Climate binning	50 × 50 equal-width grid of temperature × precipitation; cell "sampled" if at least 1 EcM sample; coverage = sampled cells / Canada cells per zone	Coverage definition adapted from Hébert et al. (2026) (Detailed methods SM1, `17_hutchinsonian.R`).
F22	Eltonian	Sample restriction (Canadian scope)	Host associations restricted to **root samples** (GenBank host/isolation_source; GlobalFungi dominant/other plant species)	Direct root extractions give strongest evidence of mycorrhizal association; bulk soil indicates only spatial proximity (Detailed methods SM1).
F23	Eltonian	Denominator	Native Canadian EcM host species list (Table 7) as denominator of potential interactions	Fixed, defensible universe of possible hosts (Detailed methods SM1).
F24	Eltonian	Matrices & metrics	Interaction matrices at SH, named-species, genus levels; report % hosts with at least 1 association, matrix fill rate, and % pairs from a single record	Multiple resolutions and a single-record fragility check (Detailed methods SM1).

#	Shortfall	Decision	Choice	Rationale
F25	Eltonian	Global scope retrieval	GenBank global query retrieves **all matching records per genus** (no cap), paginated via the NCBI history server	No truncation of heavily sequenced genera; a sensitivity check confirmed the earlier 2,000-record cap materially affected only a few macrofungal genera, so we removed the cap (`18_eltonian.R`).
F26	Sampling maps	Sampling unit (Fig S4)	One distinct raw lat/lon per record	Location-level depiction of sampling (Detailed methods SM1, Fig S4).
F27	Depth discard	Metric (Fig S3)	Per Canadian GF soil sample, % of total sequencing depth discarded by EcM-genus filtering, among samples with at least 1 EcM read	Quantifies the data cost of restricting to EcM genera (Detailed methods SM1, `20_depth_discard.R`).

1.2 PART B — Decisions outside our control (upstream datasets)

These decisions are embedded in the external datasets and tools we use. We report them as fully as the sources document them; where a source's internal choices are not public in detail, that limitation is itself noted.

1.2.1 Table B1. GlobalFungi v5 (Větrovský et al. (2020))

#	Upstream decision	What we know	Note
U1	Sample inclusion & curation	GlobalFungi aggregates published community-survey amplicon studies, re-processed through its own consistent pipeline (documented at globalfungi.com)	Studies use varied sequencing platforms/protocols; predominantly soil samples (Detailed methods SM1).
U2	SH pre-assignment	SH codes pre-assigned against a specific UNITE release (the build we pin to; see Supplemental Materials SM1)	We inherit GlobalFungi's SH assignment for its records rather than re-calling them.
U3	SH-assignment identity criterion	Documented BLASTn criterion at ~98.5% identity	We match this on the GenBank side (B5).
U4	Read processing / quality filtering	Internal to the GlobalFungi pipeline (denoising, chimera removal, abundance tabulation)	Details not re-implemented or independently audited by us.

1.2.2 Table B2. UNITE (Nilsson et al. (2019))

#	Upstream decision	What we know	Note
U5	SH clustering	Dynamic clustering thresholds define SH codes; identifiers renumbered between builds	Motivates our build-pinning (Supplemental Materials SM1).
U6	Representative sequence & taxonomy	One representative sequence and one taxonomy string per SH	We read taxonomy directly from these (Supplemental Materials SM1).
U7	Dark-taxon naming	Placeholder names ending `_sp` for unnamed lineages	We follow UNITE's convention in tie resolution (B6) and dark-fraction definition (F2).

1.2.3 Table B3. FungalTraits (Põlme et al. (2020))

#	Upstream decision	What we know	Note
U8	Trait scoring grain	Genus-level trait compilation	Constrains our EcM assignment and trait coverage to genus resolution (D3, F14).
U9	Primary-lifestyle categories	Includes `ectomycorrhizal` as a primary lifestyle	Basis of our EcM guild filter (D3).
U10	Blank vs "unknown" semantics	Blanks in some trait columns are ambiguous between "not assessed" and "trait absent"	Forces our lower-bound coverage treatment for two traits (F16).

1.2.4 Table B4. FungalRoot (Soudzilovskaia et al. (2020))

#	Upstream decision	What we know	Note
U11	Mycorrhiza-type labels	FungalRoot's occurrence-level label vocabulary (incl. ambiguous `EcM, AM undetermined`, `EcM,AM`)	Drives our qualifying/excluded-label choices (E2–E4).
U12	Genus-level recommendation (Table S2)	Genus calls based on at least 67% species-level consistency	Basis of our genus-level host route (E5).
U13	Underlying GBIF taxonomy	Occurrence records resolved against the GBIF backbone	Independent from BIEN resolvers → unresolved synonymy drop-outs (E9).

1.2.5 Table B5. BIEN / BIEN2 (Maitner et al. (2018); Moulatlet et al. (2025))

#	Upstream decision	What we know	Note
U14	Occurrence name-scrubbing	BIEN applies its own name resolution	Independent resolver → synonymy drop-outs (E9).
U15	Native Status Resolver	BIEN's algorithm classifies native vs introduced	We accept its native calls (E8).
U16	BIEN2 range models	MaxEnt-based modelled range polygons with fixed modelling settings (model name encodes settings)	We take ranges as given, reproject, and clip; we do not re-fit SDMs (E11).

1.2.6 Table B6. van Galen et al. (2025) dark-taxa data

#	Upstream decision	What we know	Note
U17	Dark-OTU definition	Proportion of EcM OTUs unassigned to species level, per GF v5 sample	Different taxonomic unit (OTU) from our SH-level dark fraction — comparison is non-equivalent (F3).
U18	Raster modelling	`percentage_dark_taxa` band from their modelled global layers	Used as-is for the Figure S1 map (`10_dark_diversity.R`).

1.2.7 Table B7. BioTIME (Dornelas et al. (2018))

#	Upstream decision	What we know	Note
U19	Study inclusion & structure	BioTIME's own criteria for what constitutes an included biodiversity time series	We filter within it (at least 1 Canadian location, at least 2 time points, fungi) but inherit its study boundaries (F9).

1.2.8 Table B8. JGI MycoCosm

#	Upstream decision	What we know	Note
U20	Genome inclusion	Which fungal genomes JGI has sequenced/portalized	Defines the Darwinian denominator; reflects JGI project priorities, not random sampling (F11).
U21	Listing completeness at retrieval	Page-copy snapshot on 5 Mar 2026 (bulk export non-functional)	Snapshot may lag the live portal; retrieval method documented (F11).

1.2.9 Table B9. WorldClim 2.1, GADM, Natural Earth, ecoregions, GBIF

#	Upstream decision	What we know	Note
U22	WorldClim interpolation	Climate surfaces interpolated from station data at 30 (Fick and Hijmans (2017))	We aggregate to ~10 (F20); interpolation error inherited.
U23	Administrative boundaries	GADM / Natural Earth boundary generalization	Affects coordinate validation and offshore snapping (D1, F18).
U24	Ecozone delineation	National Ecological Framework for Canada polygon definitions	Fixed unit for ecozone coverage (F17).
U25	GBIF specimen records	Basis-of-record labelling, georeferencing, and dataset coverage vary by publisher	We exclude geospatial-issue records but inherit publisher labelling (F4).

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